ABIBE: Adaptive Building Information-Based Extraction From Remote Sensing Imagery Using Vision–Language Models

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***Abstract*— Building extraction from remote sensing imagery is essential for urban planning, population monitoring, and emergency response. However, traditional methods often struggle with accurate building delineation due to complex architectural features and varying imaging conditions. In this article, we propose an adaptive building information-based extraction (ABIBE) framework that leverages vision–language models (VLMs) for high-precision building extraction from remote sensing imagery. Our approach introduces two key innovations: 1) a hierarchical feature transfer mechanism that selectively extracts and adapts visual representations from the JanusPro VLM and 2) a dynamic multiscale feature fusion technique that adaptively integrates these features with spatial information through cross-modal attention mechanisms. Extensive experiments on the WHU Building Dataset demonstrate that our method significantly outperforms state-of-the-art approaches, achieving 91.39% IoU and 95.50% *F*1-score. We also provide comprehensive performance analysis, including computational complexity, runtime efficiency, and resource requirements, to evaluate the practical applicability of the proposed framework. The ABIBE framework provides a promising solution for high-precision building extraction from remote sensing imagery, with particular advantages in areas with complex architectural structures and high building density.**

***Index Terms*— Building extraction, cross-modal fusion, deep learning, hierarchical feature transfer, remote sensing, semantic segmentation, vision–language models (VLMs).**

I. Introduction

ACCURATE building extraction from remote sensing imagery is essential for various applications, including urban planning, population estimation, disaster response, and environmental monitoring. The rapid advancement of remote sensing technologies has enabled the acquisition of high-resolution imagery, offering new opportunities for detailed building mapping [1], [2]. However, extracting precise building footprints remains challenging due to complex architectural structures, varying illumination conditions, shadows, occlusions, and the diversity of building styles across different geographic regions [3].

Traditional building extraction methods based on convolutional neural networks (CNNs) have made notable progress in recent years [4], [5]. These approaches typically employ encoder–decoder architectures like the U-Net [6] and the fully convolutional network (FCN) [5] to generate pixel-wise building segmentation masks. While effective for basic building identification, these methods often face limitations when dealing with complex building structures, irregular shapes, and densely packed urban environments, resulting in imprecise building boundaries and missed details.

The challenges in remote sensing building extraction arise from several factors: 1) the wide variety of building appearances, materials, and architectural styles; 2) the presence of shadows and occlusions that partially mask building features; 3) the similarity between buildings and other man-made structures; and 4) variations in image acquisition conditions such as lighting, viewing angle, and atmospheric effects. These factors make it difficult for traditional CNN-based methods to achieve consistent extraction performance across diverse scenarios.

Recent advancements in vision–language models (VLMs) have shown potential for enhancing remote sensing tasks by leveraging rich semantic representations learned from large-scale multimodal data. Models like CLIP [7] and JanusPro [8] encode visual concepts with corresponding textual descriptions, enabling cross-modal understanding that could benefit domain-specific applications. However, effectively adapting these general-purpose models to the specialized task of remote sensing building extraction presents several challenges, including the domain gap between natural images and remote sensing imagery and the need for precise pixel-level segmentation.

In this article, we propose an adaptive building information-based extraction (ABIBE), a framework that integrates vision–language features with specialized segmentation techniques for building extraction from remote sensing imagery. Our approach aims to combine the semantic understanding capabilities of pretrained VLMs with the spatial precision needed for building segmentation.

The contributions of our work include as following.

1) We develop a feature transfer approach that adapts multiscale visual representations from the JanusPro VLM for building extraction, helping to address the domain

Received 8 June 2025; accepted 7 July 2025. Date of publication 10 July 2025; date of current version 17 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62221005, in part by the Scientific and Technological Research Program of Chongqing Municipal Education Commission under Grant KJZD-K202400604, and in part by the Doctoral Research Fund of Chongqing University of Posts and Telecommunications under Grant A2023-01. *(Corresponding author: Liping Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3587692

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gap between general visual understanding and remote sensing imagery.
2) We propose a multiscale feature fusion technique that combines vision–language features with domain-specific spatial information, using text guidance to focus on relevant building characteristics.
3) We employ an enhanced loss function combining binary cross-entropy, dice, focal, and Tversky components to help address the class imbalance challenge common in building extraction tasks.
4) We provide a comprehensive analysis of performance, computational requirements, and runtime efficiency to evaluate the practical applicability of our approach for real-world deployment scenarios.
5) We conduct extensive ablation studies that isolate the contributions of hierarchical feature transfer and dynamic fusion modules, providing a detailed understanding of each component's impact on building extraction performance.

Our experimental results on both the WHU Building Dataset and INRIA Aerial Image Labeling Dataset indicate that the integration of vision–language features with specialized segmentation techniques significantly improves building extraction performance compared to conventional CNN-based approaches. The proposed ABIBE framework offers a potential solution for applications requiring detailed building maps, particularly in areas with complex architectural structures and varying urban landscapes.

The remainder of this article is organized as follows. Section II reviews related work on building extraction methods and VLMs. Section III details our proposed ABIBE framework. Section IV presents the experimental setup and evaluation results. Section V discusses the implications of our findings and potential future directions. Finally, Section VI concludes this article.

II. Related Work

A. Building Extraction From Remote Sensing Imagery

Building extraction from remote sensing imagery has evolved significantly with the advancement of deep learning techniques. Early approaches relied on traditional computer vision methods such as edge detection, morphological operations, and texture analysis [9], [10]. These methods typically require extensive manual feature engineering and parameter tuning, limiting their generalization capability across diverse geographic regions.

The introduction of deep CNNs has revolutionized building extraction by automatically learning hierarchical feature representations from data. FCNs [5] pioneered end-to-end semantic segmentation, enabling pixel-wise classification. The U-Net [6], with its encoder–decoder architecture and skip connections, has become particularly popular for remote sensing segmentation tasks due to its ability to preserve spatial information while capturing semantic context.

Several studies have extended these baseline architectures for building extraction. Liu et al. [11] proposed a multiscale CNN approach that combines features from different resolutions to improve segmentation accuracy. Tian et al. [12] introduced a pyramid pooling module to capture context at multiple scales. Huang et al. [13] developed a dense connection architecture that enhances information flow across different network layers.

More recent approaches have incorporated attention mechanisms to focus on relevant features while suppressing noise. Chang et al. [14] proposed a spatial attention module that dynamically weights feature channels based on their importance. Guo et al. [15] introduced a dual attention mechanism that captures both spatial- and channel-wise dependencies. These attention-based approaches have shown improved performance, particularly in complex urban environments with diverse building types.

The emergence of Transformer architectures has recently brought significant advancements to the field of building extraction. Wang et al. [16] introduced BuildFormer, a Vision Transformer with a dual-path structure designed specifically for building extraction that achieves state-of-the-art performance on benchmark datasets. Chen et al. [17] proposed Sparse Token Transformers that address the computational challenges of applying transformers to high-resolution remote sensing imagery while maintaining accuracy.

Despite these advancements, accurately delineating complex building structures remains a persistent challenge. Most CNN-based methods still struggle with varied architectural styles and dense urban environments due to limitations in capturing the rich semantic context necessary for distinguishing buildings from similar structures.

B. VLMs in Remote Sensing

VLMs have emerged as powerful tools for learning transferable visual representations by leveraging large-scale image–text paired data. Models like CLIP [7], Qwen2.5-VL [18], and JanusPro [8] learn joint embeddings of images and text through contrastive learning objectives, enabling zero-shot classification, cross-modal retrieval, and transfer learning to downstream tasks.

The application of VLMs to remote sensing tasks is a relatively recent development. Singha et al. [19] explored CLIP's zero-shot capabilities for remote sensing scene classification, demonstrating promising results even without domain-specific fine-tuning. Li et al. [20] proposed a contrastive VLM for zero-shot remote sensing scene classification, improving classification accuracy by learning semantic-aware visual representations and using pseudo-labeling and curriculum learning techniques for model training.

While domain-specific remote sensing models have been proposed for various tasks, general VLMs like JanusPro offer several unique advantages. These models have been trained on vast and diverse datasets spanning multiple domains, enabling them to learn rich semantic representations that capture a wide range of visual concepts and their relationships with language [21]. This broader knowledge base is particularly valuable for building extraction, as buildings exhibit tremendous diversity in architectural styles, materials, and contexts across different regions. The general-purpose training allows

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these models to develop robust feature representations that can generalize better to unseen building types and environments compared to models trained exclusively on remote sensing data.

Furthermore, VLMs encode implicit knowledge about object hierarchies [22], spatial relationships, and contextual cues through joint visual–linguistic learning. This multimodal understanding facilitates more nuanced interpretation of complex scenes, such as distinguishing buildings from similar structures based on contextual information. The ability to leverage textual descriptions also introduces flexibility in guiding the extraction process, allowing models to adapt to specific building characteristics without requiring extensive retraining.

In the remote sensing domain, Wu et al. [23] investigated the effectiveness of vision–language features for land cover segmentation, showing improvements over traditional CNN-based approaches. However, these methods primarily focus on general segmentation performance and do not specifically address the challenges of building extraction in remote sensing imagery.

Our proposed ABIBE framework bridges these research gaps by combining vision–language features from JanusPro with specialized feature fusion techniques designed for remote sensing building extraction. Unlike previous approaches, our method adaptively integrates multiscale visual representations with domain-specific features, leveraging both the semantic understanding capabilities of VLMs and the specialized requirements of building extraction.

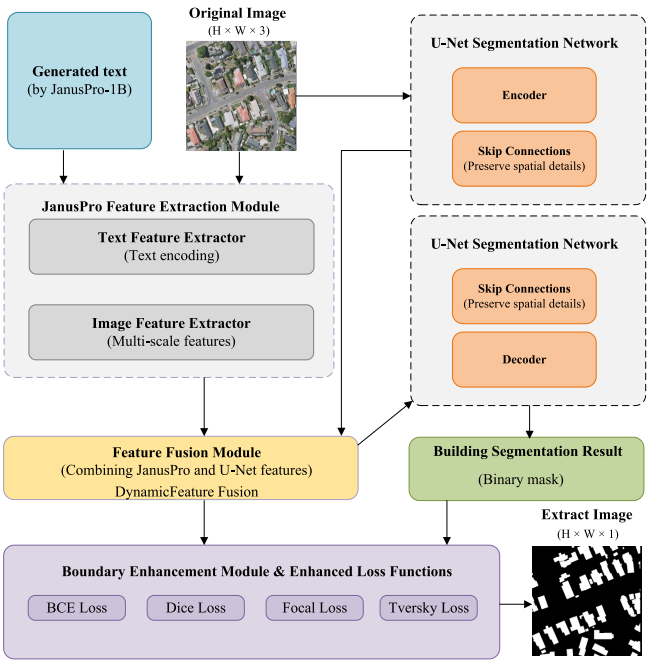


Fig. 1. Overview of the proposed ABIBE framework. The framework consists of three main components: 1) a JanusPro feature extraction module that extracts multiscale features from both original images and generated text; 2) a dynamic feature fusion module that adaptively integrates JanusPro features with U-Net features to enhance model representation capabilities; and 3) a boundary enhancement module with enhanced loss functions that combines BCE Loss, Dice Loss, Focal Loss, and Tversky Loss for precise building edge delineation. The framework processes both image inputs and optional text inputs to extract buildings with high precision.

III. Methodology

In this section, we present our ABIBE framework for high-precision building extraction from remote sensing imagery. The framework leverages vision–language features from JanusPro and enhances building segmentation through specialized fusion techniques.

A. Framework Overview

Fig. 1 illustrates the ABIBE framework. It consists of three main components: 1) a hierarchical feature transfer mechanism that extracts and adapts visual representations from JanusPro; 2) a dynamic multiscale feature fusion technique that adaptively integrates vision–language features with spatial information; and 3) a boundary enhancement module that refines building edges for precise segmentation.

Given an input remote sensing image $I \in \mathbb{R}^{H\times W\times 3}$ and an optional text description T (e.g., "aerial view of residential buildings"), our framework first extracts multiscale visual features from the JanusPro VLM. These features are then dynamically fused with domain-specific representations from the U-Net encoder. The resulting feature maps are processed by the boundary enhancement module and the U-Net decoder to produce the final building extraction result.

B. Hierarchical Feature Transfer From VLMs

The hierarchical feature transfer mechanism extracts visual representations from different layers of the JanusPro VLM and applies preprocessing and normalization techniques to adapt them for remote sensing tasks. This approach enables the integration of semantic understanding from pretrained models while addressing the feature distribution differences between natural and remote sensing imagery.

1) Terminology Clarification: To avoid confusion, we provide clear definitions of key terms used in our methodology.

1) *Multiscale Feature Aggregation:* The process of combining features extracted from different layers (3, 6, 9, and 12) of the JanusPro model, where each layer captures information at different semantic levels (low-level to high-level). This operation combines features from the *same modality* (visual) but at different scales, as described in (5).
2) *Cross-Modal Fusion:* The integration of features from *different modalities*, specifically combining visual features from JanusPro with textual features when text descriptions are available. This enables the model to leverage both visual and linguistic information through attention mechanisms, as described in (8)–(11).
3) *Adaptive Feature Fusion:* The dynamic weighted integration of aggregated JanusPro features with domain-specific features from the U-Net encoder, allowing the model to balance contributions from different feature sources based on content characteristics, as shown in (7).
4) *Feature Fusion:* A general term encompassing all fusion operations in our framework, including multiscale aggregation within modalities, cross-modal

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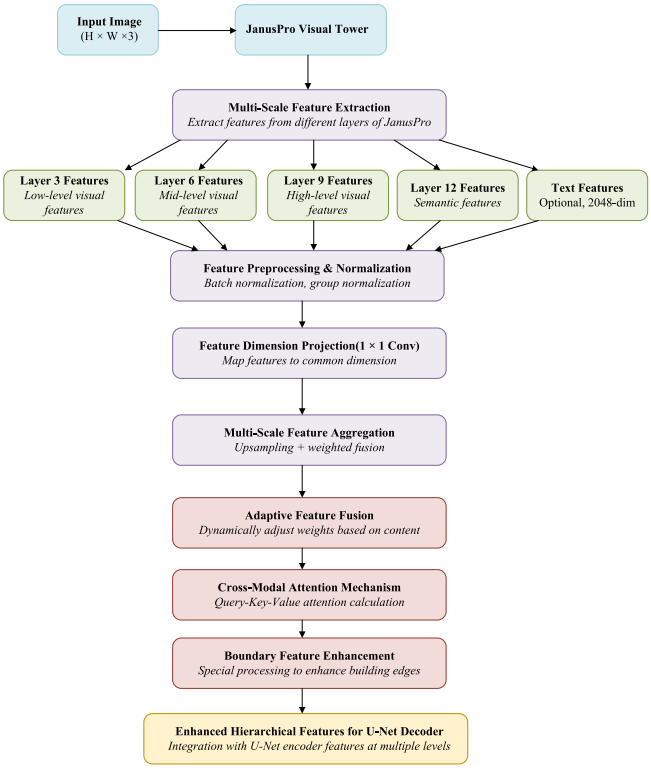


Fig. 2. Detailed workflow of the hierarchical feature extraction and processing from JanusPro. The process includes extracting multiscale features from different layers (3, 6, 9, and 12), preprocessing and normalization, dimension projection, and adaptive fusion with optional text features.

integration between modalities, and adaptive fusion with domain-specific features, resulting in enriched feature representations for building extraction.

Fig. 2 illustrates our comprehensive feature extraction and processing pipeline. The process begins by extracting features from multiple transformer layers (3, 6, 9, and 12) of the JanusPro model, which capture hierarchical visual information ranging from low-level patterns to high-level semantic concepts. These multiscale features then undergo a series of processing steps, including normalization to bridge domain gaps, dimension projection via 1×1 convolutions to achieve consistent feature spaces, and adaptive aggregation to balance contributions from different semantic levels. When text descriptions are available, they are optionally integrated through cross-modal attention mechanisms to further enhance the semantic understanding. This systematic processing creates rich, domain-adapted representations that are well-suited for the building extraction task.

2) JanusPro Feature Extraction: Given an input image I, we extract hierarchical visual features from different layers of the JanusPro vision encoder

$$F_l = \phi_l(I), \quad l \in \{3, 6, 9, 12\} \tag{1}$$

where ϕ_l represents the feature extraction function for layer l of JanusPro (with l denoting the layer index) and $F_l \in \mathbb{R}^{H_l \times W_l \times C_l}$ is the corresponding feature map. The feature dimensions vary across layers, with deeper layers having higher semantic content but lower spatial resolution.

If a text description T is provided, we also extract a text feature vector

$$F_T = \psi(T) \in \mathbb{R}^D \tag{2}$$

where ψ is the text encoder of JanusPro and D is the text embedding dimension.

The selection of JanusPro-1B as our VLM was motivated by several key considerations. While various VLMs like CLIP and Qwen2.5-VL demonstrate strong performance in cross-modal understanding, JanusPro offers unique advantages for our building extraction task. First, JanusPro-1B provides a quantized version with approximately 1B parameters, significantly smaller than most mainstream VLMs that typically exceed this parameter count even in their quantized forms. This efficient parameterization allows JanusPro to maintain high-quality visual–linguistic capabilities while substantially reducing computational and memory requirements—a critical factor for the practical deployment of building extraction systems. Additionally, compared to models like CLIP that primarily focus on global image–text alignment, JanusPro's architecture better preserves multiscale feature representations across different layers, which is particularly beneficial for our hierarchical feature transfer mechanism. The model's unified multimodal understanding also enables more effective integration of domain-specific textual prompts related to building characteristics, further enhancing the extraction precision.

3) Feature Preprocessing and Normalization: To bridge the domain gap between natural images (on which JanusPro was trained) and remote sensing imagery, we apply feature preprocessing and normalization techniques

$$\hat{F}_l = \text{Norm}(F_l) \in \mathbb{R}^{H_l \times W_l \times C_l} \tag{3}$$

where Norm represents a series of normalization operations including batch normalization, group normalization, and appropriate initialization of convolutional parameters. These operations help standardize the feature statistics for improved compatibility with remote sensing imagery characteristics.

To ensure dimensional compatibility and reduce computational complexity, we project each adapted feature map to a common embedding dimension

$$P_l = \mathcal{P}_l(\hat{F}_l) \in \mathbb{R}^{H_l \times W_l \times D'} \tag{4}$$

where $\mathcal{P}_l$ is a projection function implemented as a 1×1 convolutional layer followed by batch normalization and ReLU activation, and D' is the common embedding dimension. This projection operation performs linear dimensionality reduction to achieve consistent feature spaces across different layers.

C. Dynamic Multiscale Feature Fusion

The dynamic multiscale feature fusion module adaptively integrates features from different scales and sources, balancing semantic context with spatial details to achieve accurate building extraction.

Fig. 3 provides a detailed illustration of our feature fusion mechanism, showing how different feature sources are processed and integrated through multiple stages. This comprehensive approach enables effective multimodal feature integration for building extraction.

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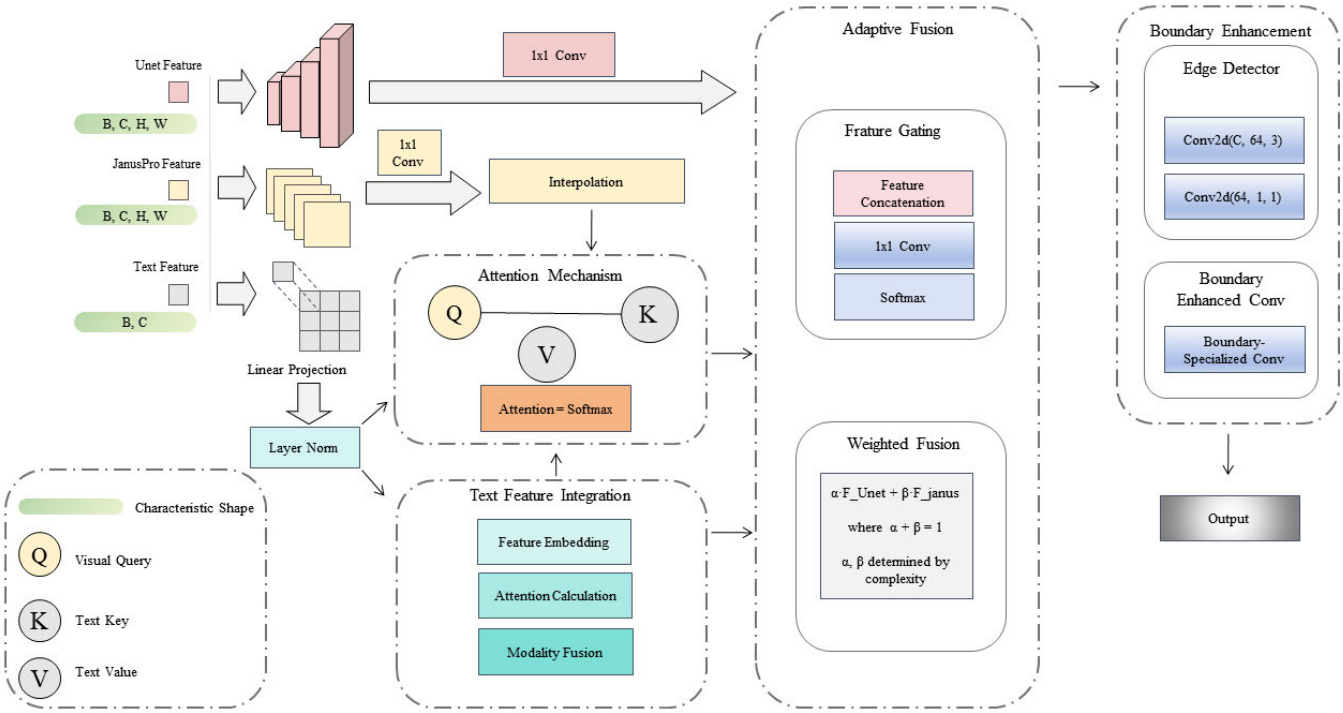


Fig. 3. Detailed architecture of the feature fusion mechanism in ABIBE. The diagram illustrates the complete feature flow from different sources (U-Net, JanusPro, and text features) through various processing stages, including attention mechanisms, feature gating, adaptive fusion, and boundary enhancement. This comprehensive fusion approach enables effective integration of multimodal information for high-precision building extraction.

1) Multiscale Feature Aggregation: We first upsample the JanusPro features from different layers to a common resolution and perform multiscale feature aggregation

$$F_{\text{agg}} = \sum_{l} w_l \cdot \text{Upsample}(P_l, H_o, W_o) \tag{5}$$

where Upsample is a bilinear upsampling operation that resizes the feature maps to the target output resolution (H_o, W_o) and w_l are learnable weights that determine the contribution of each semantic level. These weights are normalized using a Softmax function to ensure $\sum_l w_l = 1$. This multiscale aggregation combines visual features from the same modality but different semantic levels.

2) Adaptive Feature Fusion: To adaptively balance the contribution of different feature types, we employ a dynamic fusion module

$$\alpha = \sigma(f_\alpha(F_{\text{agg}}, F_T)) \tag{6}$$

where f_α is a function that computes fusion weights based on the aggregated features and text features (if available). This function is implemented as a small neural network with global average pooling and fully connected layers.

The fusion weights determine the balance between JanusPro features and domain-specific features

$$F_{\text{fused}} = \alpha \cdot F_{\text{agg}} + (1 - \alpha) \cdot F_{\text{domain}} \tag{7}$$

where F_{domain} represents domain-specific features extracted from a conventional U-Net encoder trained specifically on remote sensing building datasets. The relationship between F_{domain} and the U-Net encoder is established through direct feature extraction from corresponding encoder layers, ensuring spatial and semantic consistency. This adaptive fusion mechanism allows the model to leverage both the semantic richness of vision–language features and the domain specificity of dedicated building extraction features.

3) Cross-Modal Attention Mechanism: To further enhance the integration of visual and textual features, we implement a cross-modal attention mechanism inspired by the transformer architecture. This mechanism allows the model to focus on relevant building features guided by textual descriptions.

For the visual features F_v from JanusPro and the text features F_T, we compute

$$Q = W_q F_v, \quad K = W_k F_T, \quad V = W_v F_T \tag{8}$$

where W_q, W_k, and W_v are learnable projection matrices. The attention weights are computed as

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{9}$$

where d_k is the dimension of the key vectors. The output of the attention mechanism is

$$F_{\text{att}} = AV. \tag{10}$$

This attention-enhanced feature is then combined with the original visual features through a residual connection

$$F_{\text{enhanced}} = F_v + F_{\text{att}}. \tag{11}$$

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This cross-modal attention mechanism allows the model to selectively attend to visual features that are relevant to the textual description, improving the semantic coherence of the extracted buildings.

D. Boundary Enhancement Module

Building extraction frequently suffers from imprecise boundary delineation, especially in densely built-up urban areas. To address this challenge, we propose a dedicated boundary enhancement module that refines building edges for more accurate segmentation.

The boundary enhancement module consists of two main components: an edge detector and a boundary-specialized convolution network.

1) Edge Detection: The edge detector identifies potential building boundaries from the fused features

$$E = \mathcal{E}(F_{\text{fused}}) \tag{12}$$

where $\mathcal{E}$ is implemented as a sequence of convolutional layers with appropriate activation functions, designed to highlight edge features.

2) Boundary-Enhanced Feature Refinement: The detected edge features are then used to refine the fused features

$$F_{\text{refined}} = \mathcal{B}(F_{\text{fused}}, E) \tag{13}$$

where $\mathcal{B}$ is a boundary enhancement function that focuses on refining features near detected edges. This function is implemented as a series of dilated convolutional layers with different dilation rates to capture multiscale context around building boundaries.

The refined features are then fed into the U-Net decoder for final segmentation

$$Y_{\text{pred}} = \mathcal{D}(F_{\text{refined}}) \tag{14}$$

where $\mathcal{D}$ represents the U-Net decoder and Y_{pred} is the predicted building segmentation mask.

E. Enhanced U-Net Architecture

We integrate the fused features into an enhanced U-Net architecture for building extraction. The U-Net backbone consists of an encoder path for downsampling and a decoder path for upsampling, connected by skip connections to preserve spatial information.

1) U-Net Encoder: The U-Net encoder follows the standard structure with five levels of downsampling:

$$E_k = \mathcal{E}_k(E_{k-1}), \quad k \in \{1, 2, 3, 4, 5\} \tag{15}$$

where $E_0 = I$ is the input image and $\mathcal{E}_k$ is the encoding function at level k, consisting of two 3×3 convolutional layers followed by batch normalization and ReLU activation and a 2×2 max-pooling operation for downsampling.

2) Feature Integration: At each level of the U-Net encoder, we integrate the corresponding JanusPro features through the fusion mechanism described earlier. The integrated features are then passed to the U-Net decoder

$$\hat{E}_k = F_k^{\text{fused}} + E_k \tag{16}$$

where F_k^{fused} is the fused feature at level k, derived from the JanusPro features.

3) U-Net Decoder: The U-Net decoder upsamples the features and combines them with the corresponding encoder features through skip connections

$$D_k = \mathcal{D}_k(D_{k+1}, \hat{E}_k), \quad k \in \{4, 3, 2, 1\} \tag{17}$$

where $D_5 = \hat{E}_5$ is the bottleneck feature and $\mathcal{D}_k$ is the decoding function at level k, consisting of a 2×2 transposed convolution for upsampling, concatenation with the corresponding encoder features, and two 3×3 convolutional layers followed by batch normalization and ReLU activation.

F. Enhanced Loss Function

To address the challenges of building extraction, we propose an enhanced loss function that combines multiple loss components

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{Dice}} + \lambda_3 \mathcal{L}_{\text{Focal}} + \lambda_4 \mathcal{L}_{\text{Tversky}} \tag{18}$$

where $\mathcal{L}_{\text{BCE}}$ is the binary cross-entropy loss, $\mathcal{L}_{\text{Dice}}$ is the Dice loss, $\mathcal{L}_{\text{Focal}}$ is the focal loss, $\mathcal{L}_{\text{Tversky}}$ is the Tversky loss, and $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are weighting factors. The choice of these weights is based on extensive validation experiments comparing different weight configurations, with $\lambda_1 = 0.8$, $\lambda_2 = 1.0$, $\lambda_3 = 0.6$, and $\lambda_4 = 0.6$ providing optimal performance across different building extraction scenarios.

1) Binary Cross-Entropy Loss: The binary cross-entropy loss is defined as

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{19}$$

where N is the number of pixels, y_i is the ground truth label for pixel i, and $\hat{y}_i$ is the predicted probability.

2) Dice Loss: The Dice loss measures the overlap between the predicted and ground-truth masks

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^{N} y_i \hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon} \tag{20}$$

where ϵ is a small constant to prevent division by zero.

3) Focal Loss: The focal loss addresses class imbalance by focusing on hard examples

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{N} \sum_{i=1}^{N} \big[\alpha y_i (1 - \hat{y}_i)^{\gamma} \log(\hat{y}_i) + (1 - \alpha)(1 - y_i) \hat{y}_i^{\gamma} \log(1 - \hat{y}_i)\big] \tag{21}$$

where α is a balancing factor and γ is a focusing parameter that reduces the contribution of easy examples.

4) Tversky Loss: The Tversky loss provides flexibility in balancing false positives and false negatives

$$\mathcal{L}_{\text{Tversky}} = 1 - \frac{\text{TP} + \epsilon}{\text{TP} + \alpha \text{FP} + \beta \text{FN} + \epsilon} \tag{22}$$

where

$$\text{TP} = \sum_{i=1}^{N} y_i \hat{y}_i \tag{23}$$

$$\text{FP} = \sum_{i=1}^{N} (1 - y_i) \hat{y}_i \tag{24}$$

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$$\mathrm{FN} = \sum_{i=1}^{N} y_i(1 - \hat{y}_i) \tag{25}$$

represents true positives, false positives, and false negatives, respectively. The parameters α and β control the balance between false positives and false negatives, allowing the model to prioritize precision or recall based on specific task requirements. When $\alpha > \beta$, the model focuses more on reducing false positives, while $\alpha < \beta$ emphasizes reducing false negatives.

G. Data Preprocessing and Text Feature Generation

To ensure robust model training and consistent performance, we implement comprehensive data preprocessing techniques specifically designed for remote sensing building extraction tasks, with particular attention to handling multimodal datasets.

1) Dataset Organization and Splitting: Our preprocessing pipeline handles diverse dataset structures commonly found in remote sensing applications.

1) *Multiformat Dataset Support:* The preprocessing system supports datasets with different organizational structures, including train-only, test-only, or mixed configurations. For the INRIA Aerial Image Labeling Dataset, we implement a splitting strategy that divides the original large tiles (5000 × 5000 pixels) into smaller patches of 512 × 512 pixels.
2) *Adaptive Data Splitting:* We employ configurable splitting ratios with default settings of 70% for training, 10% for validation, and 20% for testing. The splitting process uses a fixed random seed (42) to ensure reproducible results across experiments.
3) *Image Cropping Strategy:* For large-scale imagery exceeding 1000 × 1000 pixels, we implement an intelligent cropping mechanism that divides images into nonoverlapping patches while preserving spatial relationships. Each patch maintains the target resolution of 512 × 512 pixels through bilinear interpolation when necessary.

2) Image Normalization and Standardization: All input images undergo standardized preprocessing to ensure consistent feature distributions.

1) *Pixel Value Normalization:* Images are normalized using ImageNet statistics with mean values [0.485, 0.456, 0.406] and standard deviations [0.229, 0.224, 0.225] for RGB channels, respectively. This normalization strategy aligns the remote sensing imagery statistics with the pretraining distribution of JanusPro.
2) *Resolution Standardization:* Input images are resized to 512 × 512 pixels using bilinear interpolation to maintain consistency across the dataset while preserving spatial relationships.
3) *Label Processing:* Ground-truth masks are converted to grayscale format and resized using nearest-neighbor interpolation to preserve discrete label values. Binary thresholding (threshold = 0.5) is applied to ensure clean binary masks.

3) Text Description Generation Pipeline: Our framework incorporates a systematic text generation pipeline that leverages JanusPro's multimodal capabilities.

1) *Structured Question Template:* We employ a standardized prompt template that frames the model as an expert in remote sensing: "You are an expert in the field of remote sensing architecture extraction. For the following image, answer the following questions: 1) How many buildings are visible in the image? 2) What is the shape of the buildings? 3) How are the buildings distributed in the image? Please make sure to answer each question separately and clearly. Answer each question with only one sentence."
2) *Text Cleaning and Standardization:* Generated descriptions undergo systematic cleaning to remove redundant phrases, normalize formatting, and ensure consistent three-sentence structure. A fallback mechanism provides default descriptions when generation fails.
3) *Feature Encoding:* Text descriptions are encoded into 2048-D feature vectors using JanusPro's text encoder, with mean pooling applied to obtain fixed-size representations suitable for cross-modal fusion.

4) Data Augmentation Techniques: To improve model generalization and robustness, we apply targeted augmentation strategies during training.

1) *Geometric Transformations:* Random horizontal and vertical flipping (probability 0.5), discrete rotations (90°, 180°, 270°), and synchronized transformations applied to both images and masks to maintain spatial correspondence.
2) *Photometric Augmentations:* Random brightness adjustment (±0.2) and contrast modification (±0.2) applied only to images to account for varying illumination conditions in remote sensing imagery while preserving label integrity.
3) *File Management:* The preprocessing system maintains comprehensive file tracking, supporting both copying and moving operations with optional override capabilities for iterative processing workflows.

5) Multimodal Data Validation: Our preprocessing pipeline includes robust validation mechanisms to ensure data integrity.

1) *File Correspondence Checking:* The system validates that each image has corresponding label and text feature files, using filename-based matching that handles different file extensions.
2) *Progressive Processing:* The pipeline supports resumable processing by tracking completed files and only processing remaining samples, enabling efficient handling of large datasets.
3) *Error Handling and Logging:* Comprehensive logging tracks processing statistics, failed files, and provides detailed error messages for debugging dataset issues.

H. Implementation Details

The proposed ABIBE framework is implemented using PyTorch. We use JanusPro-1B as the VLM for feature extraction, with layers 3, 6, 9, and 12 selected for hierarchical

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feature transfer. The layers were chosen based on empirical evaluation to balance semantic richness and spatial detail. The common embedding dimension D' is set to 256 for all projected features.

In the dynamic feature fusion module, the weights for multiscale feature aggregation are initialized to equal values (1/4 for each scale) and then learned during training. We perform feature standardization through batch and group normalization techniques rather than explicit domain adaptation modules. The domain-specific features are extracted from a U-Net encoder pretrained on the WHU Building Dataset.

For the enhanced loss function, we set the weighting factors $\lambda_1 = 0.8$, $\lambda_2 = 1.0$, $\lambda_3 = 0.6$, and $\lambda_4 = 0.6$ based on validation performance. The focal loss parameters are set to $\alpha = 0.25$ and $\gamma = 2.0$, while the Tversky loss parameters are set to $\alpha = 0.7$ and $\beta = 0.3$.

During training, we use the Adam optimizer with an initial learning rate of 1e-4 and a weight decay of 5e-4. The learning rate is reduced by a factor of 0.5 when validation performance plateaus for 5 epochs. We employ early stopping with a patience of 20 epochs to prevent overfitting.

Data augmentation techniques, including random flipping, rotation, scaling, and color jittering, are applied during training to improve model generalization. For text features, we use a set of building-related prompts such as "aerial view of buildings," "satellite image of urban structures," and "buildings with complex roof patterns."

IV. Experiments

A. Experimental Setup

1) Datasets: We evaluate our proposed method on two widely used building extraction datasets.

1) *WHU Building Dataset: [24]* It is a high-resolution (0.3 m) aerial imagery dataset containing building footprints across urban, suburban, and rural areas. The dataset consists of 8189 buildings across 4736 image tiles of size 512 × 512 pixels, with a standard train/val/test split of 4736/1036/2416 tiles.
2) *INRIA Aerial Image Labeling Dataset:* It covers diverse urban landscapes across several cities in Europe and the United States with a spatial resolution of 0.3 m. It contains 360 tiles of size 5000 × 5000 pixels, split into training and testing sets (180 tiles each). For our experiments, we crop these tiles into smaller patches of 512 × 512 pixels. The final training set consists of 10 125 images, the validation set consists of 1458 images, and the test set consists of 2997 images.

The ground-truth building masks for both datasets are provided as binary images, where pixels belonging to buildings are labeled as 1, and background pixels are labeled as 0.

2) Evaluation Metrics: We employ the following metrics to evaluate the performance of building extraction.

1) *Intersection Over Union (IoU)*

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{26}$$

where P is the predicted building mask, G is the ground-truth mask, TP is true positives, FP is false positives, and FN is false negatives.

2) *Precision*

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{27}$$

3) *Recall*

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{28}$$

4) *F1-Score:* The harmonic mean of precision and recall

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}}. \tag{29}$$

3) Compared Methods: We compare our ABIBE framework with several state-of-the-art methods.

1) *U-Net [6]:* A classic encoder–decoder architecture with skip connections widely used for semantic segmentation tasks.
2) *U-Net*:* Our enhanced version of U-Net with improved encoder–decoder connections and attention mechanisms.
3) *DeepLabv3+ [25]:* An advanced semantic segmentation model that combines atrous spatial pyramid pooling with an encoder–decoder structure.
4) *FCN [26]:* An FCN for semantic segmentation.
5) *PSPNet [27]:* A network that utilizes pyramid pooling modules to aggregate context information at different scales.
6) *ViT [28]:* A Vision Transformer model adapted for semantic segmentation tasks.
7) *TransUNet [29]:* A hybrid Transformer–CNN model designed for medical image segmentation, adapted for remote sensing.
8) *MSB-Net [30]:* A multiscale building extraction network with attention mechanisms.
9) *EGAFNet [31]:* A specialized network for building extraction using edge guidance and adaptive fusion.
10) *AGs-Unet [32]:* An attention-gated U-Net designed for building extraction.
11) *Easy-Net [33]:* A lightweight building extraction network specifically designed for remote sensing imagery.

We also include ablation variants of our framework.

1) **+janus:* Incorporates JanusPro visual features with the enhanced U-Net.
2) **+text:* Uses only text features with the enhanced U-Net.
3) ABIBE: Our complete framework integrating both visual and textual features with the enhanced U-Net.

4) Implementation Details: All experiments were conducted on Ubuntu 18.04.5 LTS, using NVIDIA L40S GPUs with 48 GB of memory. This hardware configuration provided sufficient computational resources for model training and inference.

The performance metrics and visualization results presented in this article are from the Linux platform. All models were implemented in PyTorch and trained for 150 epochs with

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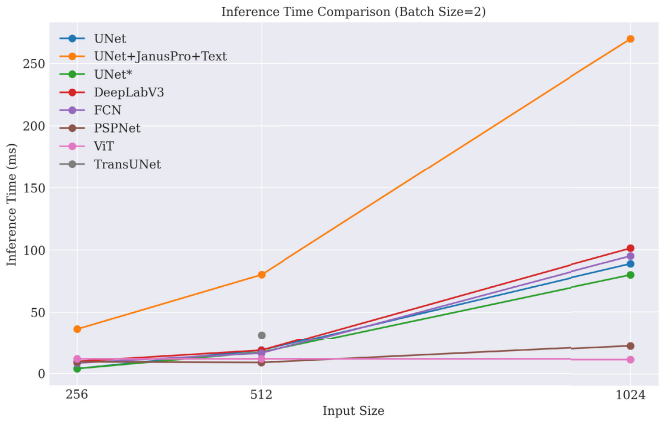


Fig. 4. Inference time comparison of different models across various input sizes with batch size 2. The ABIBE framework (UNet + JanusPro + Text) requires significantly more processing time than traditional CNN-based approaches, particularly for larger input images.

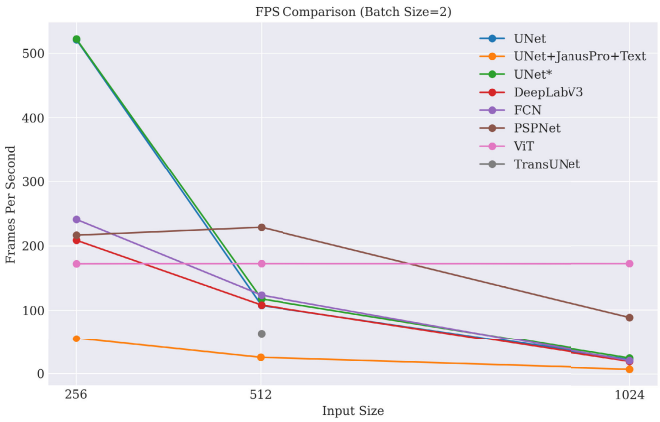


Fig. 5. FPS comparison across different models and input sizes with batch size 2. While ABIBE demonstrates lower throughput than traditional methods, it still maintains practical frame rates for many offline applications.

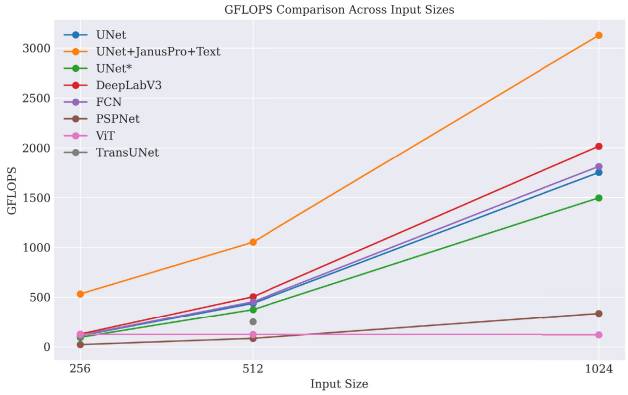


Fig. 6. GFLOPS comparison across different models and input sizes. ABIBE has substantially higher computational requirements, scaling quadratically with input resolution.

a batch size of 2. We used the Adam optimizer with an initial learning rate of 1e-4, which was reduced by a factor of 0.5 when validation performance plateaued for 10 epochs.

For text descriptions in our ABIBE framework, we used a standardized prompt template that frames the model as an expert in remote sensing architecture extraction. The template instructs the model to answer three specific questions about the input image: 1) How many buildings are visible in the image? 2) What is the shape of the buildings? and 3) How are the buildings distributed in the image? The prompt explicitly requests that each question be answered separately and clearly, with only one sentence per question. This structured approach helps guide the VLM to focus on specific building characteristics that are relevant for the extraction task, while ensuring consistent text features during both training and inference.

Data augmentation techniques included random horizontal and vertical flipping (probability 0.5), random rotation ($\pm 30°$), random scaling (0.8–1.2), and random brightness and contrast adjustments (± 0.2). These augmentations were applied consistently across all compared methods to ensure fair evaluation.

B. Computational Complexity and Performance Analysis

Before presenting the segmentation results, we first analyze the computational requirements of our proposed framework compared to other methods. This analysis is crucial for understanding the practical applicability of the models in real-world scenarios.

Fig. 4 shows the inference time comparison across different models for various input sizes. As expected, the ABIBE framework requires significantly more processing time than conventional CNN-based approaches due to the incorporation of vision–language features. For a 512×512 input image with batch size 2, ABIBE requires approximately 82 ms per inference compared to 24 ms for the baseline U-Net, representing a $3.4\times$ increase in processing time. This computational overhead becomes more pronounced for larger input sizes, with ABIBE requiring 274 ms per inference for 1024×1024 images compared to 91 ms for U-Net.

Fig. 5 presents the frames per second (FPS) comparison, which is the inverse of inference time. For 512×512 images, ABIBE achieves approximately 12 frames/s compared to 120 frames/s for the baseline U-Net. While this represents a significant reduction in throughput, it remains practical for many offline remote sensing applications where real-time processing is not a strict requirement.

Fig. 6 shows the GFLOPS (billion floating-point operations per second) comparison, which measures the theoretical computational complexity of each model. ABIBE requires approximately 1050 GFLOPS for processing a 512×512 image, which is $2.5\times$ higher than DeepLabV3+ (420 GFLOPS) and $2.7\times$ higher than U-Net (390 GFLOPS). This computational demand increases quadratically with input resolution, reaching over 3100 GFLOPS for 1024×1024 images.

Similarly, Fig. 7 presents the GMACs (billion multiply–accumulate operations) comparison, which focuses specifically on matrix multiplication operations that dominate deep learning workloads. The pattern closely resembles the GFLOPS comparison, with ABIBE requiring significantly more matrix operations than other models.

To better understand the real-world training performance, Fig. 8 presents the average batch processing time for different models on the INRIA dataset. The standard UNet exhibits the highest batch processing time at 0.207 s per training batch,

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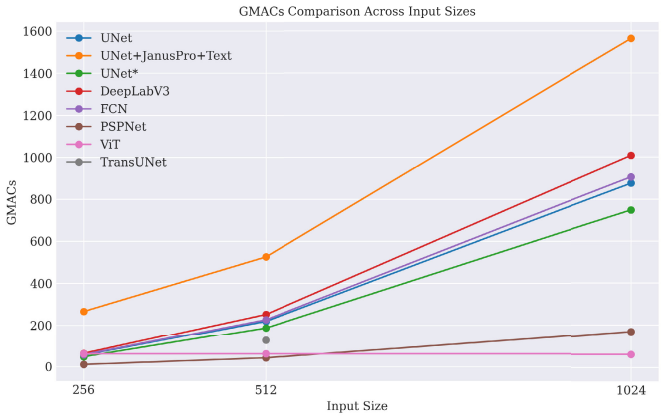


Fig. 7. GMACs comparison across different models and input sizes. The pattern is similar to GFLOPS, with ABIBE showing significantly higher matrix multiplication operations due to the vision–language feature processing.

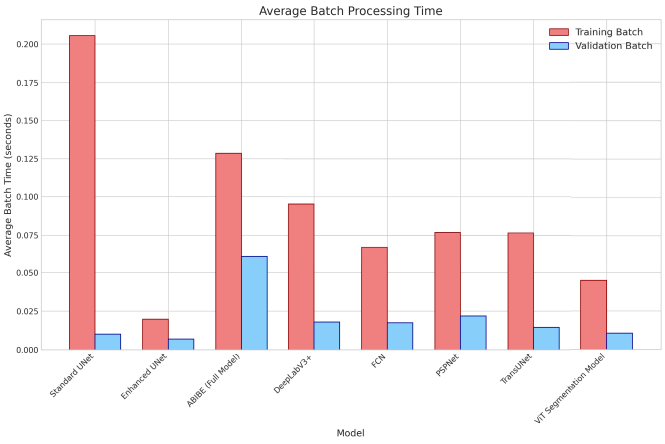


Fig. 8. Average batch processing time for different models on the INRIA dataset. The comparison shows training and validation batch times for each model, highlighting the significantly higher computational cost of Standard UNet and the ABIBE framework compared to enhanced UNet and other models. The ABIBE model requires approximately 0.129 s per training batch, while the validation batch processing is more efficient at 0.062 s.

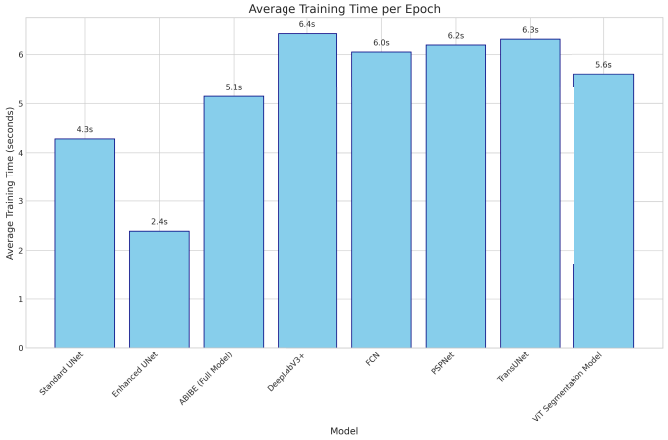


Fig. 9. Average training time per epoch for different models on the INRIA dataset. DeepLabV3+ requires the longest training time (6.4 s/epoch), while enhanced UNet is the most efficient (2.4 s/epoch). The ABIBE framework requires 5.1 s/epoch, positioning it in the middle range of computational efficiency.

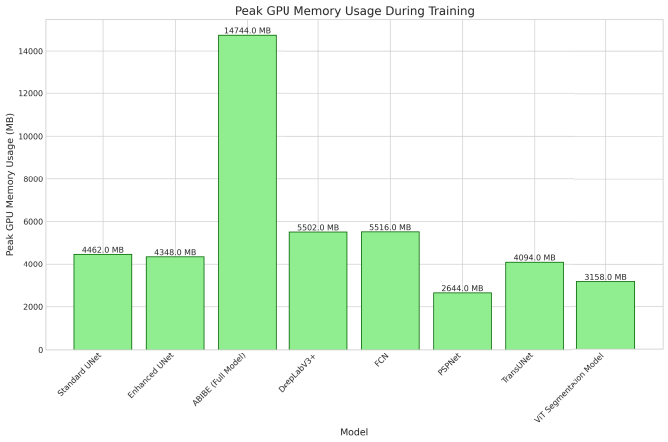


Fig. 10. Peak GPU memory usage during training for different models on the INRIA dataset. The ABIBE framework demands significantly more memory (14 744 MB) compared to other models, highlighting its resource-intensive nature. The PSPNet demonstrates the most memory-efficient architecture at 2644 MB.

while our enhanced UNet significantly improves efficiency with only 0.020 s per batch. The ABIBE framework (full model) requires 0.129 s per training batch, which is higher than most models but still more efficient than the standard UNet. Interestingly, while training times show significant variation, validation times are more consistent across models, with ABIBE requiring 0.062 s per validation batch. This difference between training and validation times highlights the impact of backpropagation and gradient computation during the training phase.

Fig. 9 illustrates the average training time per epoch for each model on the INRIA dataset. DeepLabV3+ exhibits the longest training time at 6.4 s/epoch, followed closely by TransUNet at 6.3 s. The enhanced UNet demonstrates exceptional efficiency at only 2.4 s/epoch, which is 2.1× faster than our ABIBE framework (5.1 s/epoch). This epoch-level analysis provides insights into the overall training efficiency, which is particularly important for applications requiring frequent model updates or training with limited computational resources.

Fig. 10 presents the peak GPU memory usage during training for each model. The ABIBE framework exhibits substantially higher memory requirements at 14 744 MB, which is approximately 3.3× higher than the standard UNet (4462 MB) and 2.7× higher than DeepLabV3+ (5502 MB). This significant memory footprint is primarily due to the large VLM being loaded into GPU memory during training. The PSPNet demonstrates the most memory-efficient architecture at 2644 MB, while the ViT Segmentation model also shows good memory efficiency at 3158 MB. These memory requirements are crucial considerations for deployment scenarios where GPU resources may be limited or shared across multiple tasks.

Table I presents the model parameter counts and model sizes for different methods. ABIBE has a total of 2148.27 million parameters, which is approximately 22× larger than TransUNet (98.39 million) and 69× larger than the baseline U-Net (31.04 million). However, it is important to note that most of these parameters come from the frozen JanusPro backbone. The number of trainable parameters for ABIBE is

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TABLE I
MODEL PARAMETER COUNT AND SIZE COMPARISON

Method	Total(MB)	Trainable(MB)
U-Net	31.04	31.04
U-Net*	28.95	28.95
PSPNet	48.40	48.40
DeepLabV3+	60.99	60.99
FCN	54.30	54.30
ViT	29.19	29.19
TransUNet	98.39	98.39
ABIBE	2148.27	59.04

TABLE II
CORE COMPONENT ABLATION STUDY ON THE WHU BUILDING DATASET

Method	IoU(%)	Precision(%)	Recall(%)	F1(%)
U-Net	87.17	86.61	85.23	85.42
U-Net*	89.11	94.13	94.36	94.24
*+janus	90.82	95.45	94.93	95.19
*+text	90.72	95.55	94.73	95.13
ABIBE (Ours)	**91.39**	**96.20**	**94.81**	**95.50**

59.04 million, which is comparable to DeepLabV3+ (60.99 million) and slightly larger than FCN (54.30 million). This distinction is crucial for understanding the training requirements versus inference requirements of the model.

The model size analysis reveals that ABIBE requires approximately 4.6 GB of storage, which is significantly larger than other methods. This increased storage requirement primarily results from the inclusion of the JanusPro VLM. While this larger storage footprint may pose challenges for deployment on resource-constrained edge devices, it remains feasible for server-based or cloud applications where storage capacity is less restricted.

Collectively, these analyses demonstrate that while the ABIBE framework offers superior segmentation performance, it comes with considerable computational and memory requirements. For applications where accuracy is paramount and computational resources are available, ABIBE provides a compelling solution. However, for deployment scenarios with strict computational constraints, simpler models like enhanced UNet or PSPNet may offer better efficiency-accuracy tradeoffs.

C. Results on the WHU Building Dataset

Table II presents a comprehensive ablation study on the WHU Building Dataset, demonstrating the effectiveness of our approach and the contribution of each component in our ABIBE framework.

Our ABIBE framework achieves an IoU of 91.39%, outperforming the baseline U-Net by 4.22% points and the enhanced U-Net (U-Net*) by 2.28% points. The improvement is consistent across all metrics, with our method achieving a precision of 96.20%, a recall of 94.81%, and an $F1$-score of 95.50%.

The ablation studies in Table II demonstrate the contribution of each component in our framework. The enhanced U-Net (U-Net*) provides substantial improvements over the standard U-Net, achieving an IoU improvement of 1.94% points (89.11% versus 87.17%). When incorporating JanusPro visual features (*+janus), we observe a further improvement to 90.82% IoU, indicating the value of VLM features for this task. Using only text features (*+text) also shows a significant improvement over U-Net* with an IoU of 90.72%, which is slightly lower than using visual features alone.

TABLE III
LOSS FUNCTION ABLATION STUDY ON THE WHU BUILDING DATASET

Loss Configuration	IoU(%)	F1(%)
ABIBE (Multi-loss)	**91.39**	**95.50**
BCE Only	91.24	95.42
Baseline (Multi-loss)	90.98	95.27
BCE + Dice	90.84	95.20
BCE + Dice + Focal	90.67	95.11
Dice Only	90.62	95.08

TABLE IV
MODULE ABLATION STUDY ON THE WHU BUILDING DATASET

Configuration	IoU(%)	Relative to ABIBE
ABIBE (Full Model)	**91.39**	baseline
w/o Multi-scale Fusion	91.28	-0.11
w/o Text Features	90.91	-0.48
Only Dynamic Fusion	90.91	-0.48
w/o Boundary Enhancement	90.28	-1.11
Only Hierarchical Features	88.66	-2.73
Enhanced U-Net (U-Net*)	89.11	-2.28
Base U-Net	87.17	-4.22

Our complete ABIBE framework, which integrates both visual and textual features from JanusPro with adaptive fusion techniques, achieves the best performance across all metrics. This confirms the effectiveness of our hierarchical feature transfer mechanism and dynamic multiscale feature fusion approach.

Table III shows the results of our loss function ablation study. Our complete ABIBE framework with the multicomponent loss function achieves the best performance (91.39% IoU), outperforming simpler loss configurations, including BCE-only (91.24% IoU). This validates the effectiveness of our enhanced loss function design that combines multiple objectives to handle the complexity of building extraction tasks.

Table IV presents a detailed module ablation study that isolates the contributions of different components in our ABIBE framework. The results demonstrate the following.

1) *Multiscale Fusion:* Removing multiscale fusion causes a minimal decrease (-0.11% IoU), indicating that other components effectively capture multiscale information.
2) *Text Features:* Disabling text features results in a -0.48% IoU decrease, confirming the value of textual guidance for building extraction.
3) *Boundary Enhancement:* Removing boundary enhancement causes a -1.11% IoU decrease, highlighting its importance for precise building delineation.
4) *Hierarchical Features:* Using only hierarchical features results in a significant -2.73% IoU decrease, demonstrating the complementary nature of different feature types.
5) *Dynamic Fusion:* Using only dynamic fusion achieves similar performance to removing text features, indicating the interconnected nature of these components.

Table V presents a comparison with additional state-of-the-art methods. Our ABIBE framework outperforms all compared

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TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE WHU BUILDING DATASET

Method	IoU(%)	Pre(%)	Recall(%)	F1(%)
Easy-Net	89.16	94.94	93.48	94.20
AGs-Unet	85.50	93.70	–	–
PSPNet	73.10	72.55	74.44	73.01
DeepLabV3+	86.24	85.09	85.10	84.42
FCN	80.59	81.45	81.08	80.88
ViT	81.07	82.92	82.23	82.09
TransUNet	84.79	85.61	84.10	84.41
MSB-Net	89.13	94.25	94.49	94.01
EGAFNet	89.30	93.82	**94.88**	94.35
ABIBE (Ours)	**91.39**	**96.20**	94.81	**95.50**

TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE INRIA AERIAL IMAGE LABELING DATASET

Method	IoU(%)	Pre(%)	Recall(%)	F1(%)
U-Net	73.70	84.11	78.79	80.81
PSPNet	70.24	74.88	73.35	73.69
DeepLabV3+	77.19	81.07	77.77	78.95
FCN	74.19	78.98	76.32	77.20
ViT	69.65	77.09	73.15	74.42
TransUNet	75.21	79.13	75.33	77.65
ABIBE (Ours)	**78.07**	**88.55**	**86.83**	**87.68**

methods, including specialized architectural designs like the Easy-Net and EGAFNet. Notably, our approach achieves a significant improvement in precision (96.20%) compared to other methods, indicating more accurate building boundary detection and fewer false positives. While the EGAFNet achieves slightly higher recall (94.88% versus 94.81%), our method provides better overall performance in terms of IoU and $F1$-score.

D. Results on INRIA Aerial Image Labeling Dataset

To evaluate the generalization capability of our framework across different geographic regions and building styles, we conducted experiments on the INRIA Aerial Image Labeling Dataset, which contains diverse urban landscapes from multiple cities.

Table VI shows the results on the INRIA dataset. Our ABIBE framework significantly outperforms all compared methods, achieving an IoU of 78.07%, which is 0.88% points higher than the second-best method (DeepLabV3+ with 77.19%). The performance gap is even more substantial compared to traditional CNN-based approaches like the U-Net (4.37% points improvement) and PSPNet (7.83% points improvement).

These results demonstrate the superior generalization capability of our approach across different datasets and geographic regions. The integration of vision–language features from JanusPro provides a rich semantic understanding that helps the model adapt to diverse building styles and urban layouts. This adaptability is particularly valuable for large-scale building extraction applications that need to process imagery from multiple geographic regions.

E. Qualitative Results

Fig. 11 presents qualitative results on the WHU Building Dataset, comparing our ABIBE framework with state-of-the-art methods. The visual comparison demonstrates the superior segmentation capability of our approach, particularly in challenging scenarios with complex building structures, closely spaced buildings, and varying illumination conditions.

The PSPNet and FCN generally capture the overall building shapes but struggle with precise segmentation, often producing fragmented or merged building regions. DeepLabV3+ achieves better segmentation accuracy but still shows noticeable errors in complex regions. In contrast, our ABIBE framework produces clean and accurate building segmentation, effectively capturing fine details and preserving building shapes even in challenging cases.

Fig. 12 provides a visual comparison of our ablation studies, illustrating the contribution of each component in our framework. The enhanced U-Net (U-Net*) significantly improves upon the baseline U-Net, producing cleaner boundaries and fewer fragmented regions. The addition of JanusPro visual features (*+janus) further enhances the segmentation quality, particularly in complex areas with multiple adjacent buildings. The text features (*+text) contribute to better contextual understanding, helping to disambiguate building regions from similar structures. Finally, our full ABIBE framework, combining all components, achieves the most accurate segmentation with well-preserved building shapes and clear boundaries.

The qualitative results highlight the effectiveness of our feature fusion approach, which leverages both semantic context from JanusPro and spatial details from the U-Net backbone. The adaptive fusion mechanism further improves the model's ability to focus on relevant building features, resulting in more accurate segmentation.

V. DISCUSSION

A. Analysis of Key Findings

Our experimental results reveal several important insights about building extraction in remote sensing imagery and the

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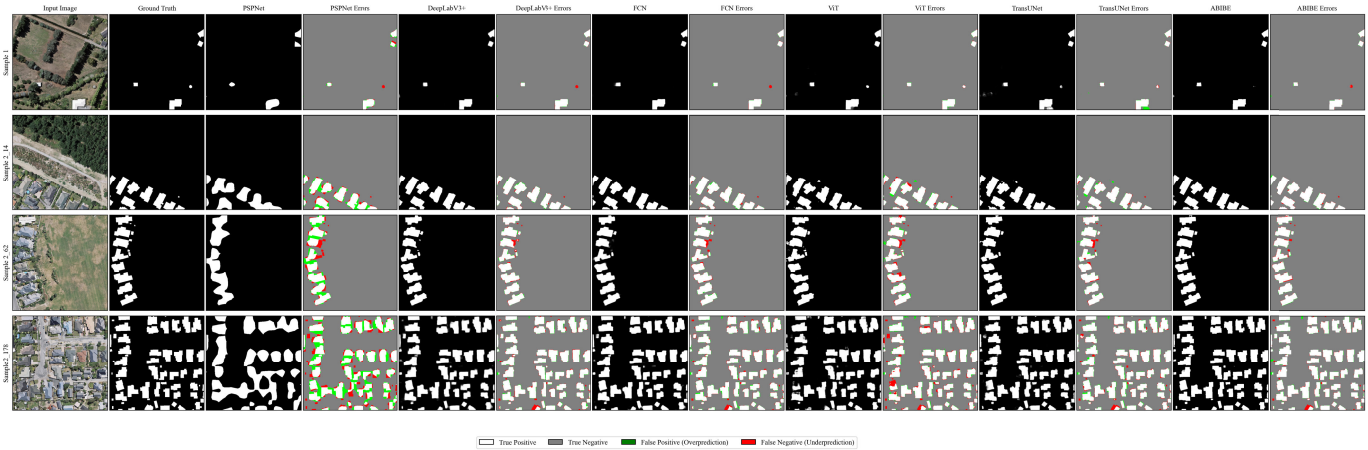


Fig. 11. Qualitative comparison of building extraction results on the WHU Building Dataset. From left to right: input image, ground truth, PSPNet, PSPNet Errors, DeepLabV3+, DeepLabV3+ Errors, FCN, FCN Errors, Vit, Vit Errors, TransUNet, TransUNet Errors, and our ABIBE framework with corresponding error visualization. The error maps use color coding where white represents true positives, gray shows true negatives, green indicates false positives (overprediction), and red shows false negatives (underprediction). Our method achieves more accurate building segmentation with fewer errors, particularly in areas with complex structures and closely spaced buildings, as evidenced by the reduced red and green areas in the error maps.

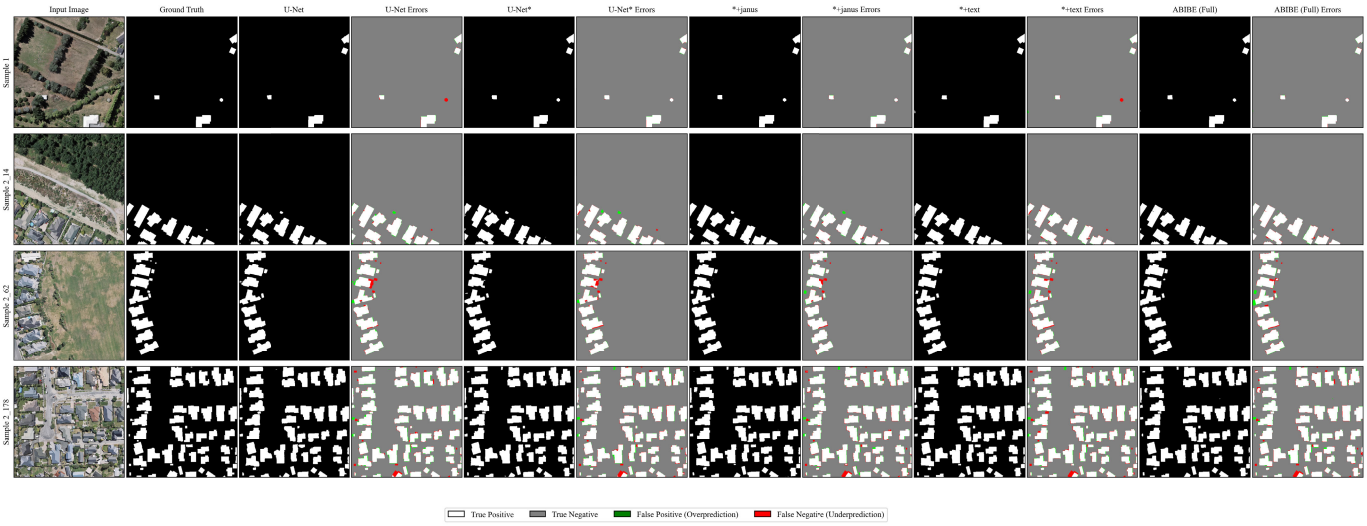


Fig. 12. Visual comparison of ablation studies on the WHU Building dataset. From left to right: input image, ground truth, U-Net, U-Net Errors, U-Net*, U-Net* Errors, *+janus, *+janus Errors, *+text, *+text Errors, and our full ABIBE framework with visualization of errors. The error maps highlight true positives (white), true negatives (gray), false positives/overprediction (green), and false negatives/underprediction (red). The images demonstrate the incremental improvements achieved by each component, with the full ABIBE framework producing the most accurate building segmentation with fewer errors, particularly in complex urban environments.

effectiveness of our proposed approach. First, the incorporation of vision–language features from JanusPro significantly enhances building extraction performance. The comparison between the enhanced U-Net (U-Net*) and its variants with JanusPro features (*+janus) shows a clear improvement in IoU and $F1$-score, demonstrating the value of rich semantic representations learned from large-scale image–text paired data. These features provide a strong foundation for understanding building characteristics across diverse architectural styles and imaging conditions.

The hierarchical feature transfer mechanism in our ABIBE framework effectively leverages these vision–language features by selectively adapting multiscale representations to the building extraction task. The ablation results confirm that integrating JanusPro visual features provides substantial improvements in IoU (from 89.11% to 90.82%) and $F1$-score (from 94.24% to 95.19%). This suggests that effective domain adaptation and hierarchical integration of vision–language features are crucial factors in bridging the gap between general visual understanding and domain-specific building extraction tasks.

The integration of text guidance provides consistent improvements in building extraction performance. The ablation studies indicate that combining visual and textual features leads to the best overall performance (IoU of 91.39% and $F1$ score of 95.50%). This demonstrates that targeted textual guidance can help the model disambiguate complex visual features and attend to specific architectural elements.

B. Loss Function and Module Contribution Analysis

An important finding from our comprehensive ablation study is the superior performance of our multicomponent loss

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function (91.39% IoU) compared to simpler loss configurations, including BCE-only (91.24% IoU). This result validates the effectiveness of combining multiple loss objectives for handling the complexity of building extraction tasks. The multicomponent approach provides robustness across different building extraction scenarios and effectively handles class imbalance, pixel-level accuracy, region overlap optimization, and precision–recall tradeoffs simultaneously.

The module ablation study reveals the interconnected nature of our framework components. While individual modules like multiscale fusion show relatively small isolated contributions (-0.11% IoU when removed), the combination of all components produces synergistic effects that significantly outperform simpler configurations. The boundary enhancement module shows the largest individual contribution (-1.11% IoU when removed), highlighting its importance for precise building delineation in complex urban environments.

Notably, the hierarchical feature transfer mechanism contributes substantially to performance, with the "only hierarchical features" configuration achieving 88.66% IoU compared to the enhanced U-Net baseline of 89.11% IoU. This demonstrates that vision–language features can provide competitive performance even when used in isolation, validating the effectiveness of our feature adaptation approach.

C. Generalization and Practical Implications

The cross-dataset evaluation on the INRIA dataset further confirms the generalization capability of our approach across different geographic regions and building styles. The significant performance improvement over state-of-the-art methods (78.07% IoU compared to 77.19% for the second-best method) demonstrates the adaptability of our framework to diverse urban landscapes, which is particularly valuable for large-scale building extraction applications.

The qualitative results visually confirm the superior segmentation capability of our approach, particularly in challenging scenarios. The clean and accurate building segmentation achieved by our ABIBE framework, with well-preserved building shapes and clear boundaries, demonstrates the effectiveness of our feature fusion approach in leveraging both semantic context and spatial details. The visual comparison of ablation studies further illustrates the incremental improvements achieved by each component, with the full ABIBE framework producing the most accurate building segmentation.

D. Computational Efficiency and Practical Considerations

While our ABIBE framework achieves superior segmentation performance, it comes with increased computational requirements that warrant careful consideration for practical applications. The performance analysis reveals that our approach requires significantly more processing time and computational resources compared to traditional CNN-based methods.

The inference time analysis shows that ABIBE requires approximately $3.4\times$ more processing time than the baseline U-Net for 512×512 images, with this gap widening for larger input sizes. Similarly, the GFLOPS and GMACs comparisons indicate substantially higher computational complexity, with ABIBE requiring $2.5\times$–$3\times$ more operations than conventional approaches.

These increased computational demands translate to higher hardware requirements for deployment. The model size analysis shows that ABIBE requires approximately 4.6 GB of storage, primarily due to the inclusion of the JanusPro VLM. While this larger footprint may pose challenges for deployment on resource-constrained edge devices, it remains feasible for server-based or cloud applications where storage and computational resources are less restricted.

Despite these increased requirements, there are several practical considerations that may mitigate these challenges.

1) The number of trainable parameters in ABIBE (59.04 million) is comparable to other advanced segmentation models like DeepLabV3+ (60.99 million), meaning that the training process is not significantly more resource-intensive than existing approaches.
2) The inference speed, while slower than traditional methods, still maintains practical frame rates for many offline remote sensing applications. For a 512×512 image, ABIBE achieves approximately 12 frames/s, which is sufficient for batch processing of satellite or aerial imagery where real-time performance is not a strict requirement.
3) The framework's architecture allows for potential optimizations such as model quantization, knowledge distillation, or hardware-specific acceleration that could improve computational efficiency without significantly sacrificing accuracy.
4) For applications where computational resources are severely constrained, the ablation variants (*+janus or *+text) offer reasonable tradeoffs between performance and efficiency, providing substantial improvements over baseline methods with lower computational requirements than the full ABIBE framework.

E. Advantages of General VLMs

Our results substantiate the advantages of using general VLMs like JanusPro for remote sensing building extraction. Unlike domain-specific remote sensing models that are trained solely on aerial or satellite imagery, JanusPro has been exposed to diverse visual concepts and their textual descriptions across multiple domains. This broad knowledge base enables it to develop rich semantic representations that capture complex relationships between visual patterns and linguistic concepts.

The strong performance of JanusPro features in our experiments confirms the value of this general knowledge for building extraction tasks. The model's exposure to various architectural styles, materials, and contexts during pretraining enables it to recognize building structures even in unfamiliar environments, leading to better generalization capabilities. This is particularly valuable for remote sensing applications, where building styles can vary significantly across different geographic regions.

Furthermore, the ability to leverage textual descriptions through JanusPro's multimodal understanding provides

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additional flexibility in guiding the extraction process. This text-visual integration allows our framework to focus on specific building characteristics based on textual prompts, without requiring extensive retraining or architectural modifications. The improvement observed with the full ABIBE framework demonstrates how language can effectively complement visual features for remote sensing tasks.

The hierarchical nature of features extracted from different layers of JanusPro also contributes to the effectiveness of our approach. These multiscale representations capture both low-level visual patterns and high-level semantic concepts, providing a comprehensive understanding of building structures and their relationships with the surrounding environment. By selectively adapting and fusing these features with domain-specific representations, our framework effectively bridges the gap between general visual understanding and specialized building extraction requirements.

F. Limitations and Future Work

Despite the promising results, our ABIBE framework has several limitations that warrant further investigation. The computational complexity of the model increases significantly when incorporating vision–language features, which may limit its applicability in resource-constrained environments or real-time applications. Future work could explore more efficient feature extraction and fusion techniques to reduce computational requirements while maintaining performance.

The current implementation relies on JanusPro as the VLM for feature extraction. Exploring the integration of other VLMs could provide valuable insights into the transferability of different pretrained representations to the building extraction task. Future research could develop a unified framework that leverages the complementary strengths of multiple VLMs, potentially through ensemble learning or knowledge distillation approaches.

While our approach shows strong performance on the datasets used in this study, it may still face challenges in extreme environmental conditions or with very unusual architectural structures not represented in the training data. Incorporating additional contextual information, such as geographic location, climate conditions, or cultural background, could further enhance the model's ability to adapt to diverse scenarios.

Our current approach focuses on single-image building extraction, but many real-world applications involve temporal data or multiple imagery sources. Extending our framework to incorporate temporal information or multisource data could further improve building extraction performance, particularly in challenging scenarios such as seasonal variations, partial occlusions, or adverse imaging conditions.

The principles and techniques developed in our ABIBE framework could be extended to other remote sensing tasks, such as road network extraction, land cover classification, and change detection. Developing task-specific variations of our framework represents a promising direction for future research.

VI. Conclusion

In this article, we present ABIBE, a framework that leverages VLMs for high-precision building extraction from remote sensing imagery. Our approach addresses the challenges of building extraction by introducing two key innovations: a hierarchical feature transfer mechanism that adapts JanusPro visual representations and a dynamic multiscale feature fusion technique that adaptively integrates these features with spatial information through cross-modal attention mechanisms.

Extensive experiments on the WHU Building Dataset and INRIA Aerial Image Labeling Dataset demonstrate that our ABIBE framework significantly outperforms state-of-the-art methods, achieving 91.39% IoU and 95.50% $F1$-score on WHU, and 78.07% IoU and 87.68% $F1$-score on INRIA. The comprehensive ablation studies confirm the importance of each component in our framework, with detailed analysis isolating the contributions of hierarchical feature transfer and dynamic fusion modules. Our loss function ablation study validates the effectiveness of the multicomponent loss approach, which outperforms simpler loss configurations and provides optimal performance for building extraction tasks.

Qualitative results further demonstrate the superior segmentation capability of our approach, particularly in challenging scenarios with complex building structures and varying illumination conditions. The module ablation study provides a detailed understanding of each component's impact, showing that the boundary enhancement module contributes significantly to precise building delineation, while the hierarchical feature transfer mechanism effectively adapts vision–language features for domain-specific tasks.

We also conduct a comprehensive analysis of computational complexity, runtime efficiency, and resource requirements to evaluate the practical applicability of our framework. While ABIBE requires more computational resources than traditional methods, it maintains practical inference speeds for many remote sensing applications and offers several potential optimization pathways for deployment in resource-constrained environments.

The proposed ABIBE framework represents a significant advancement in remote sensing building extraction, providing a promising solution for applications requiring high-fidelity building maps such as urban planning, population estimation, and disaster response. The integration of VLMs with adaptive fusion techniques provides a powerful approach for leveraging general visual understanding in domain-specific tasks. Future work will focus on exploring more efficient implementations, incorporating additional contextual information, and extending the approach to other remote sensing tasks.

Code Availability

The implementation code of the ABIBE framework is available at https://github.com/Yongt-D/ABIBE. The repository currently contains essential information about the framework structure. The complete codebase with documentation will be made publicly available under the MIT license upon acceptance of this article.

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ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions to improve the quality of this article.

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